

INDEX NUMBER.....

SIGNATURE.....

KAAF UNIVERSITY COLLEGE



SCHOOL OF NURSING AND MIDWIFERY

DEPARTMENT OF NURSING

END OF FIRST SEMESTER EXAMINATION- 2022/2023 ACADEMIC YEAR

KNUR: 317 MICROBIOLOGY (3 CREDIT HOURS)

TIME: 8AM DURATION: 1:40MINUTES

DATE: 11th DECEMBER, 2022

INSTRUCTION FOR THE PAPER:

1. Read the instructions of each section and answer accordingly
2. Answer ***ALL*** Questions
3. Do not leave your seat unless you are given permission by an Invigilator
4. Do not communicate in any way with any other candidate in the examination room.
5. All section A on the question paper and section B must be written in the answer book(s) provided.
6. Circle the alphabet corresponding to the most appropriate answer
7. PDA's and mobile phones are NOT permitted in this examination.
8. You are allowed to use your own calculators, pen, pencils and rulers.
9. You are not to leave the examination room until after the first hour of the examination.
10. An announcement will be made when there are 30 minutes of the exam time left.
11. At the end of the examination, remain seated until your answer book(s) have been collected and you have been told you to leave.

GOOD LUCK

Examiner : Seth Selassie Dzah

Instructions: answer all questions. Circle the alphabet corresponding to the most appropriate answer.

1. An individual who harbours a disease-causing agent but does **not** show signs and symptoms is called
A) **carrier.**
B) definitive host .
C) susceptible host.
D) portal of exit.
2. Alcohol-based hand rub should **not** be used when
A) **hands are visibly soiled.**
B) nurses have busy schedule.
C) nurses prefer to use it.
D) there is no water.
3. A healthy individual who is at risk of infection is called
A) carrier.
B) definitive host .
C) **susceptible host.**
D) portal of exit.
4. Transfer of DNA from one bacterium to another by virus is termed
A) bacteriophage.
B) conjugation.
C) **transduction.**
D) transformation.
5. Typhoid fever is a disease caused by bacteria. Which of the following is the correct way of writing the scientific name of these bacteria?
A) **Salmonella Typhii**
B) *Salmonella typhii*
C) Salmonella typhii
D) salmonella typhii
6. Which of the following is a bacterial adhesin?
A) Cell wall
B) Cytosol

- C) Pilli
- D) Plasmid
7. Which of the following is NOT a predisposition to cholera?
- A) Adequate clean drinking water
- B) Marine shell fish consumption.
- C) Poverty
- D) Poor sanitation
8. Which of the following surface structures in bacteria is utilized in cell-cell attachment?
- A) Cilia
- B) Flagella
- C) Pilli
- D) Spinae
9. Bacillus anthracis is Gram positive. It will stain at the end of the procedure.
- A) colorless
- B) pink
- C) purple
- D) red
10. Bacteria cells have smaller ribosomes compared to those of eukaryotic cells. The advantage of this is the production of antibiotics to specifically attack ribosomes.
- A) 100s and not 90s
- B) 80s and not 70s
- C) 50s and not the 60s
- D) 90s and not 100s
11. Bacterial endospores help.....
- A) aerobic bacteria to survive high oxygen tension.
- B) anaerobic bacteria to survive high oxygen tension.
- C) anaerobic bacteria to survive low oxygen tension.
- D) facultative bacteria to survive low oxygen tension.
12. Effective hand washing should be done
- A) by nurses at least three times a shift.

- B) only after procedures.
- C) with medicated soap and water.
- D) with soap and running water.
13. Exotoxins are
- A) lowly antigenic.
- B) highly toxic.
- C) coded by genes on DNA.
- D) heat stable.
14. Master Johnson refused to wash his hands after visiting the toilet. His argument is that he uses the water closet alone and cleans it with disinfectants regularly, hence no risk of infection. Which principle of disease prevention is likely to be violated?
- A) Change in normal flora location can lead to disease causation.
- B) Hand washing after visiting the toilet is mandatory.
- C) Modification of normal flora composition of a site can result in diseases.
- D) Water closets always inhabit pathogenic microbes.
15. Media formulated for the culture of fastidious bacteria is referred to as
- A) basic
- B) differential
- C) enriched
- D) selective
16. Nurse Ativor reads culture result for urine, which reported to contain gram positive bacteria. The nurse is worried because
- A) normal flora of the urinary system are gram negative.
- B) the identified bacteria might be difficult to treat.
- C) urine is expected to be sterile.
- D) normal flora of the urinary system are gram positive.
17. The stationary exhibits
- A) adapt to the growth medium.
- B) destroy themselves in the medium.
- C) equilibrium between death and replication.

- D) multiply in the growth media.
18. Staphylococcus aureus is normal flora of the
- A) colon.
 - B) hand.
 - C) nose.
 - D) throat.
19. The genome in bacteria can be found in the
- A) cytoplasm.
 - B) nucleoid region.
 - C) nucleus.
 - D) periplasm.
20. Master Boadu has been rushed to your unit, with a history of severe vomiting and diarrhoea. He is very weak; and has been diagnosed with cholera. The nurse should immediately
- A) administer intravenous infusions.
 - B) serve oral rehydration salt.
 - C) serve prescribed antibiotics.
 - D) teach patient personal hygiene practices.
21. Bacteria may acquire new genes by taking up naked DNA from the environment. This is called
- A) bacteriophage.
 - B) conjugation.
 - C) transduction.
 - D) transformation.
22. Which of the following is the correct way of writing the scientific name of the bacteria which cause anthrax?
- A) *Bacillus anthracis*
 - B) *Bacillus anthracis*
 - C) Bacillus Anthracis
 - D) bacillus anthracis
23. Which of the following structures facilitates movement in bacteria?
- A) Cell wall

INDEX NUMBER.....

SIGNATURE.....

- B) Cytosol
- C) Flagellum
- D) Pilli

24. Which of the following is a characteristic of exotoxins?

- A) Gene is present in the DNA
- B) Low toxicity
- C) Polypeptide in composition
- D) Poor antigenicity

25. Which of the following processes reduces the degree of virulence of bacteria?

- A) Attenuation
- B) Deactivation
- C) Hibernation
- D) Inactivation

26. Which of the following surface structures is utilised by bacteria during conjugation?

- A) Cilia
- B) Flagella
- C) Pilli
- D) Spinae

27. Gram negative bacteria stain at the end of the gram stain procedure.

- A) colourless
- B) pink
- C) purple
- D) red

28. During therapy some antibiotics attack ribosomes.

- A) 100s and not 90s
- B) 80s and not 70s
- C) 50s and not the 60s
- D) 90s and not 100s

29. When spores are formed, it helps

- A) aerobic bacteria to survive high oxygen tension.

INDEX NUMBER.....

SIGNATURE.....

- B) anaerobic bacteria to survive high oxygen tension.
 - C) anaerobic bacteria to survive low oxygen tension.
 - D) facultative bacteria to survive high oxygen tension.
30. All the following are characteristics of endotoxins **except**:
- A) lowly antigenic.
 - B) **heat labile**.
 - C) coded by genes on DNA.
 - D) heat stable.
31. In the gram staining procedure, iodine plays the role of
- A) catalyst.
 - B) **mordant**.
 - C) primary stain.
 - D) secondary stain.
32. Which of the following media can be used to grow some bacteria while inhibiting others?
- A) basic
 - B) differential
 - C) enriched
 - D) **selective**
33. Bacteria which infect the urethra
- A) **are gram negative**.
 - B) might be difficult to treat.
 - C) possess capsules.
 - D) are gram positive.
34. During the lag phase of bacteria growth, they
- A) **adapt to the growth medium**.
 - B) hibernate.
 - C) die in the medium.
 - D) multiply in the growth media.
35. The tetanus toxoid vaccine contains ...
- A) capsule polysaccharide.

- B) live attenuated organism.
 - C) preformed antibodies.
 - D) **modified toxin.**
36. The stationary phase of bacteria growth curve exhibits
- A) adapt to the medium.
 - B) death of bacteria.
 - C) **equilibrium between death and replication.**
 - D) rapid multiplication of bacteria.
37. The decline phase of bacteria growth enables them
- A) adapt to the growth medium.
 - B) destroy themselves in the medium.
 - C) **die in the medium.**
 - D) multiply in the growth media.
38. Which of the following nursing interventions is **most important** in the management a wound likely to be infected by anaerobic bacteria?
- A) Administering prescribed antibiotics.
 - B) **Debriding the wound and aerate it.**
 - C) Dressing the wound with normal saline.
 - D) Teaching the patient how to keep the wound dry.

INSTRUCTION: Use the scenario below to answer questions 39 to 41

*Master Boateng reported to your unit with a clean shallow wound on the right leg.
The nurse perceives risk for infection by bacteria.*

39. Which class of bacteria would the nurse be LEAST worried about?
- A) Aerobic
 - B) Anaerobic
 - C) **Facultative**
 - D) Obligate aerobe
40. Which of the following nursing interventions is NOT indicated to prevent the perceived infection?

- A) Administering prescribed antibiotics.
 - B) Debriding the wound and aerate it.
 - C) Dressing the wound with normal saline.
 - D) Teaching the patient how to keep the wound dry.
41. Which criterion of classification did the nurse consider?
- A) Gram reaction
 - B) Location in wounds
 - C) Oxygen utilization
 - D) Shape
42. What is the approximate size of the bacterial cell?
- A. 0.5 to 1.0 micrometer in diameter
 - B. 1mm in diameter
 - C. 2 micrometer in diameter
 - D. 2mm in diameter
43. Which among the following is a “Spirochete”?
- A. *Escherichia coli*
 - B. *Mycobacterium tuberculosis*
 - C. *Staph. aureus*
 - D. *Treponema pallidum*
44. Which of the following microorganism is eliminated in canned foods?
- A. *Clostridium botulinum*
 - B. *Coxiella burnetii*
 - C. *Lactobacillus*
 - D. *Mycobacterium tuberculosis*
45. Lipopolysaccharide in cell walls is characteristic of?
- A. Algae
 - B. Fungi
 - C. Gram-negative bacteria
 - D. Gram-positive bacteria
46. Growth of microbes in a solid media is identified by the formation of?
- A. Colonies
 - B. Pellicle at the top of media
 - C. Sediment at the bottom
 - D. Turbidity
47. In the classification of bacteria according to shape, which among the following refers to

cluster arrangement of bacterial cells?

- A. *Sarcinae*
- B. *Staphylococci*
- C. *Streptococci*
- D. *Tetrads*

48. Ribosomes of eukaryotes have a sedimentation coefficient of?

- A. 50S
- B. 70S
- C. 80S
- D. 90S

49. The isolation of *Neisseria gonorrhoeae* by the use of certain antibiotics in media is an example of:

- A. assay media.
- B. differential media.
- C. enriched media.
- D. selective media.

50. The Reactive Oxygen Species (ROS) produced by some bacteria are degraded by which of the following enzymes?

- A. Catalase
- B. Lyase
- C. Peroxidase
- D. Superoxide dismutase, Catalase and Peroxidase

51. Which of the following ROS is a powerful oxidizing agent?

- A. Hydrogen peroxide
- B. Hydroxyl radical
- C. Singlet oxygen
- D. Superoxide radical

52. The portion of the growth curve where rapid growth of bacteria is observed is known as:

- A. decline phase.
- B. lag phase.
- C. logarithmic phase.
- D. stationary phase.

53. Lag phase is also known as:

- A. generation time.
- B. period of initial adjustment.
- C. period of rapid growth.
- D. transitional period.

54. Which phase shows the reproduction rate approximately equal to the death rate?

- A. Death phase
- B. Lag phase
- C. Log phase
- D. Stationary phase

55. Syphilis is caused by which of the following organism?

- A. *Treponema carateum*
- B. *Treponema endemicum*
- C. *Treponema pallidum*
- D. *Treponema pertenue*

56. Reduction of virulence is known as:

- A. attenuation.
- B. cure.
- C. exaltation.
- D. treatment.

57. The bacterial cells are at their metabolic peak during phase.

- A. decline
- B. lag
- C. log
- D. stationary

58. Endotoxin produced by gram negative bacteria is present in:

- A. inner membrane.
- B. Lipopolysaccharide.
- C. Peptidoglycan.
- D. theichoic acid.

59. The anthrax disease is most frequently infected from:

- A. cattle
- B. sheep
- C. rats
- D. both a and b

60. Staining material of gram-positive bacteria is:

- A. crystal violet
- B. fast green
- C. haematoxylon
- D. safranin

61. Staining material of gram-negative bacteria is:

- A. crystal violet

- B. fast green
 - C. haematoxylon
 - D. safranin
62. Mordant used in Gram staining is:
- A. crystal violet
 - B. iodine
 - C. safranin
 - D. all of the above
63. A facultative anaerobic:
- A. only grow anaerobically
 - B. only grow in the presence of O_2
 - C. ordinarily an aerobe but can grow in absence of O_2
 - D. ordinarily an anaerobe but can grow with O_2
64. Exotoxins are:
- A. heat labile
 - B. heat stable
 - C. part of cell wall
 - D. polymerized complexes
65. The differences between Gram-positive and Gram-negative bacteria is shown to reside in the:
- A. cell membrane
 - B. cell wall
 - C. mesosomes
 - D. nucleus
66. Rod shaped bacteria are known as:
- A. bacilli
 - B. cocci
 - C. comma forms
 - D. pleomorphic forms
67. Gram positive bacteria appear as with Gram stain.
- A. Pink
 - B. Violet
 - C. Black
 - D. Yellow
68. Attenuation means:
- A. inactivation of bacteria
 - B. killing of the bacteria
 - C. more activating the bacteria

- D. both 1 and 2
69. Main cause for Cholera is:
A. mosquitoes
B. poverty and insanitation
C. toxin produced by pesticides
D. none of the above
70. Gas gangrene bacillus is:
A. facultative aerobe
B. facultative anaerobe
C. obligate aerobe
D. obligate anaerobe
71. The technique used to avoid all microorganisms is:
A. disinfection
B. dusting
C. sterilization
D. washing
72. The technique used to use instrument surfaces safe to touch is:
A. disinfection
B. dusting
C. sterilization
D. washing
73. Which of the following methods will have no effect on spores?
A. autoclaving
B. drying
C. hot air oven
D. None of the above
74. Mycobacterium tuberculosis is:
A. facultative aerobe
B. facultative anaerobe
C. obligate aerobe
D. obligate anaerobe
75. Quinolones would:
A. destroy cell membrane of bacteria cells leading to their death
B. increase water retention inside bacteria cells hence swelling
C. inhibit nucleic acid synthesis in bacteria cells.
D. inhibit transpeptidase.
76. In a case of severe diarrhea, which of the following interventions will be of least priority to you?

INDEX NUMBER.....

SIGNATURE.....

- A. Administration of analgesics.
- B. Administration of antibiotics.
- C. Administration of IV fluids.
- D. Education on the disease condition.

77. The genome in bacteria can be found in the:

- A. cytoplasm.
- B. nucleoid region.
- C. nucleus.
- D. periplasm.

78. To treat patients, ciprofloxacin is prescribed. The nurse understands that ciprofloxacin

- A. destroys cell membrane in bacteria.
- B. inhibits bacterial dna synthesis.
- C. inhibits cell wall synthesis in bacteria.
- D. targets 30s ribosomal subunit.

INSTRUCTION: answer true or false to the questions below by circling the letter which corresponds to the most appropriate answer.

79. The greater percentage of the cell wall of Gram-positive bacteria is made up of peptidoglycan.

A) True

B) False

80. The organisms that possess hyaluronidase, collagenase and coagulase will normally cause infection by invasion.

A) True

B) False

81. The time laps in the log phase of the infectious pathogen is the only determinant of the incubation period of an infection.

A) True

B) False

82. Administration of anti-tetanus serum is crucial in the management of all wounds.

A) True

B) False

83. The acid-fast bacilli are very slow growers with a dividing time of 6weeks due the presence of mycolic acid in their cell wall.

A) True

B) False

84. Anti-tetanus serum is an active vaccine.

A) True

B) False

85. *Bacillus anthracis*, a spore forming organism, could be controlled by burying the carcass of an infected dead animal.

A) True

B) False

86. Bacteria spores are destroyed by disinfection.

A) True

B) False

87. During the stationary phase of the bacterial growth curve binary fission is equal to the death rate.

A) True

B) False

88. Pseudomembranous colitis is a condition caused by antibiotic abuse of which *Clostridium difficile* is implicated.

A) True

B) False

89. The bacterial growth medium that allows only some bacteria to grow on culture is appropriate for the growth of fastidious organisms.

A) True

B) False

90. The organisms that possess hyaluronidase, collagenase and coagulase can cause cellulitis.

A) True

B) False

91. The period when bacterial growth is stationary due to acclimatization in a new environment is the stationary phase.

A) True

B) False

92. The presence of a slime layer is an important pathogenic feature in bacteria causing dental infections.

A) True

B) False

INDEX NUMBER.....

SIGNATURE.....

93. Administration of tetanus toxoid is crucial in the management of all wounds.
A) True
B) False
94. *Staph. aureus* is a fast dividing organism due to its short dividing time.
A) True
B) False
95. The meningitis serum is an active vaccine.
A) True
B) False
96. *Bacillus anthracis*, a spore forming organism, could be controlled by burning the carcass of an infected dead animal.
A) True
B) False
97. During the stationary phase of the bacterial growth curve binary fission is equal to the death rate.
A) True
B) False
98. Isolated patient should be cared for with separate personal protective equipment.
A) True
B) False
99. Antibiotic abuse predisposes to candidiasis.
A) True
B) False
100. Bacteria can attain antibiotic resistance through enzyme production.
A) True
B) False